

CAREER PROFILE

- Embedded Linux engineer with ~9 years taking connected products from board bring-up through secure, field-updatable deployment.
- Yocto distributions, BSP and kernel work, CI/CD, and hardware-in-the-loop test.
- Product cybersecurity (EN 18031 / EU RED, secure boot, SBOM) and OTA update systems.
- Electrical Engineering graduate with honors distinction.

EXPERIENCES

Senior Software Engineer

04.2024 - PRESENT

PTx, AGCO's Precision Agriculture Technology business

- Develop embedded Linux platforms for agricultural equipment electronics; working knowledge of CAN bus and protocol.
- Applied EN 18031 (EU RED cybersecurity) to a connected product: asset identification, decision-tree assessment, and evidence for third-party certification; tracking CRA readiness.
- Implemented secure boot and signed firmware images; performed threat modeling and key management backed by a hardware security element.
- Generate and maintain SBOMs for a Yocto-based product.
- Implemented RAUC OTA updates across multiple A/B and asymmetric partition layouts: U-Boot integration, secure boot fallback handling, recovery from failed kernel, rootfs, and application updates, and the tests to prove it.
- Build and deliver containerized applications that run on embedded targets.
- Integrated AI-assisted development into daily embedded workflow for code generation, review, test authoring, and documentation, with defined boundaries on where generated code is not trusted without verification.

Senior Software Engineer

12.2021 - 04.2024

ESW Team at Appareo Systems (acquired by AGCO, 01.2022)

- Performed board bring-up for custom PCB with heterogeneous SOC (IMX8DXL), debugging with JTAG/SWD, GDB, logic analyzer, and oscilloscope.
- Integrated drivers for cellular modems, WiFi, and bluetooth.
- Setup dockerized Jenkins with manual and GitOps centric builds, using Gitflow with automated build and deployment of releases.
- Developed and utilized HIL testing with Labgrid on RPI4.
- Generated SDKs and docker images for quick cross-compilation of C++ Applications.

Software Engineer 1

05.2017 - 12.2021

System Team at Vaddio

- Working with FPGA and python developers to create an integrated embedded system that routes audio and video signals in professionally built AV systems.
- I help to build a custom Linux distribution from the 'I' up with the Yocto Project.
- Our team uses many other tools during development such as Jenkins, Jira, Git, Bitbake, and Puppet.

Software Engineer

11.2018 - 12.2021

Dojo Five

- Develop embedded software for Particle, Nordic, and STM32 microcontrollers
- Utilized custom hardware with the Particle platform for a custom door access solution
- Developed with NRF52 SDK to create devices in a complex robotic system with BLE services

Embedded Linux and OTA Experiments

ONGOING

Personal projects, self-directed

- Compared Mender and OSTree against RAUC to understand A/B, atomic, and rollback-safe update strategies, and authored a custom update agent to work through the trade-offs first hand.
- Managed device fleets with Balena: provisioning, remote access, and delivery of containerized applications to embedded targets.

PROJECTS

Hardware Integrated Prototyping Environment at SDSU Engineering Expo - Nathan Genetzky, Jordan Ulmer, Tanner Johnson (SDSU, 2017)

Design and Verification of a SPI to JTAG Interface Adapter - Nathan Genetzky, Jordan Ulmer (SDSU, 2017)

Interactive User Interface with PIC18 Microcontroller - Nathan Genetzky, Drake Jen0 (SDSU, 2017)

Particle Projects - Firmware for micro controllers sold by particle.io for use with wifi or cellular cloud devices.

SKILLS & PROFICIENCY

Yocto (Build System, Embedded Distro, BSP Support)

Developer Tools (Docker, Jenkins, Git, GDB, JTAG/SWD)

Terminal Workflow (Vim, GNU tools, shell scripting)

Linux Kernel (Drivers, Config, BSP)

Python Middleware (DBus, REST, GObject, threading)

FPGA Development (SW Interface, Reusable Verilog, Automated Builds)



**Nathan
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Software and Hardware

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EDUCATION

BS Electrical Engineering

South Dakota State
University
2012 - 2017

Minor Software Engineering

South Dakota State
University
2012 - 2017

LANGUAGES

Bash (~10 years)

Python (~8 years)

C++ (~9 years)

C (~7 years)

Dockerfile (~5 years)

Java for Android (~1 year)

INTERESTS

hobby-electronics, home
automation, embedded-linux

ice and rock climbing

snowboarding and skiing

gardening, cooking

jetskiing, boating, water
skiing

camping, kayaking, volleyball